

EX.NO:8 INTER -THREAD communication

Program:

```
class Railway {
    int seats = 2;

    synchronized void bookTicket() {
        try {
            while (seats == 0) {
                System.out.println("No seats available. Booking thread is waiting...");
                wait();
            }
            seats--;
            System.out.println("Ticket booked successfully.");
            System.out.println("Available seats: " + seats);
        } catch (InterruptedException e) {
            System.out.println(e);
        }
    }

    synchronized void cancelTicket() {
        seats++;
        System.out.println("Ticket cancelled successfully.");
        System.out.println("Available seats: " + seats);
        notifyAll();
    }
}

class Booking extends Thread {
    Railway railway;
    Booking(Railway railway) {
        this.railway = railway;
    }
    public void run() {
        for (int i = 0; i < 3; i++) {
            railway.bookTicket();
            try {
                Thread.sleep(1000);
            } catch (InterruptedException e) {
                System.out.println(e);
            }
        }
    }
}
```

```

    }
}
}
}

```

```

class Cancellation extends Thread {
    Railway railway;
    Cancellation(Railway railway) {
        this.railway = railway;
    }
    public void run() {
        try {
            Thread.sleep(2000);
        } catch (InterruptedException e) {}
        for (int i = 0; i < 2; i++) {
            railway.cancelTicket();
            try {
                Thread.sleep(1000);
            } catch (InterruptedException e) {
                System.out.println(e);
            }
        }
    }
}

```

```

public class RailwayBooking {
    public static void main(String[] args) {
        Railway railway = new Railway();
        Booking booking = new Booking(railway);
        Cancellation cancellation = new Cancellation(railway);
        booking.start();
        cancellation.start();
    }
}

```

Output:

```

Ticket booked successfully.
Available seats: 1
Ticket booked successfully.
Available seats: 0

```

No seats available. Booking thread is waiting...

Ticket cancelled successfully.

Available seats: 1

Ticket booked successfully.

Available seats: 0